Glass ceilings or glass doors? The role of firms in male-female wage disparities

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Abstract. I use Canadian linked employer-employee data to examine whether women face a glass ceiling in the labour market. I also measure the extent to which the glass ceiling comes about because women are segregated into lower-paying firms, or because they are segregated into lower-paying jobs within firms. I find clear evidence that women experience a glass ceiling that is driven by their disproportionate sorting across firm types (glass doors) rather than within firms. I find no evidence that these results are supply-driven. However, my results are consistent with predictions of an efficiency wage model where high-paying firms discriminate against females.

Résumé. Plafonds de verre ou portes de verre? Le rôle des firmes dans les disparités de salaires entre hommes et femmes. L'auteur utilise des données canadiennes sur les employeurs et les employés pour supputer si les femmes font face à un plafond de verre sur le marché du travail. Il jauge jusqu'à quel point ce plafond de verre est attribuable au fait que les femmes sont déportées vers des firmes à bas salaires ou vers des emplois à bas salaires dans les firmes. Il semble que l'expérience du plafond de verre des femmes soit attribuable au fait que les femmes se trouvent plutôt distribuées différemment entre les types de firmes (le cas de portes de verre) qu'entre les types d'emplois à l'intérieur des firmes. On ne trouve pas d'évidence que ce soit un phénomène engendré du côté de l'offre. Cependant ces résultats sont compatibles avec les prédictions du modèle des salaires d'efficience où les firmes à hauts salaires discriminent contre les femmes.

JEL classification: J16, J31

1. Introduction

In Canada as in many countries, women earn lower wages than men. Despite modest improvement in the last two decades, there still exists a considerable gap in wages between men and women in Canada (Baker and Drolet 2010). Designing

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effective policies to improve women's labour market outcomes requires knowledge about the mechanisms that underlie this persistent gender wage gap. Previous studies of the gender wage gap have focused mainly on gender differences in conditional mean wages. While this comparison is interesting, it is indicative only of the wage outcomes of average workers in each group. More recently, there has been a growing interest in examining the gender wage gap throughout the wage distribution. This has been motivated mainly by a popular notion that females face a glass ceiling in the labour market; that is, females are under-represented (over-represented) in high (low) wage regions of the wage distribution, and their under-representation becomes more pronounced as we move up the wage distribution. As a consequence, the gap between male and female wages will be larger at the top of the wage distribution than at the middle or bottom.

Developing policies that effectively address potential barriers blocking the advancement of women in the labour market requires knowledge about the magnitude of any glass ceiling and about the underlying mechanisms that give rise to it. For instance, if the glass ceiling is driven by barriers to employment at high-paying firms, then policies such as employment equity policies that target employment decisions directly will be more effective. However, if the glass ceiling is driven by gender wage disparities within firms, then policies such as pay equity policies that target wages directly and policies that promote gender equality in promotion opportunities will be more effective.

This study estimates the gender wage gap at different points of the conditional wage distribution in order to examine whether females in Canada face a glass ceiling. I also extend the previous empirical literature on glass ceilings faced by women by quantifying the extent to which the observed economy-wide glass ceiling is driven by female segregation into low-paying firms (defined as a *glass door effect*) versus female segregation into low-paying jobs within firms (defined as a *within-firm glass ceiling effect*).[1]

Finally, I investigate some of the potential underlying sources of inter-firm wage differentials and inter-firm gender segregation. I empirically test the implications of two competing theoretical models—the compensating differentials model and the efficiency wage model with gender discrimination. The results help to illuminate some of the underlying mechanisms that generate the observed difference in male-female sorting across high-paying and low-paying firms.

I find clear evidence that females face an economy-wide glass ceiling: the gender wage gap increases from -0.13 log points at the 10th percentile of the conditional wage distribution to -0.23 log points at the 90th percentile. I find strong evidence that the economy-wide glass ceiling is driven mainly by glass doors. Two thirds of the increase in the gender wage gap between the 10th and 90th percentile of the conditional wage distribution is due to differential sorting of males and females across high- and low-paying firms (i.e., glass doors). I find no evidence that inter-firm wage differences and inter-firm gender differences in sorting can be

1 To the best of my knowledge, the term glass door was first used by Pendakur and Woodcock (2010) in the context of native-immigrant wage differentials.

accounted for by compensating differentials. However, my results are consistent with predictions of an efficiency wage model where high-paying firms discriminate against females. My results also suggest that after controlling for observables and inter-firm sorting, females still experience a sizeable within-firm wage gap throughout the wage distribution.

The remainder of this article is organized as follows. Section 2 reviews the previous literature. Section 3 describes the empirical methodology, including quantifying the glass door effect, and describes my data set. Section 4 provides the estimation results, and finally, section 5 concludes. The online appendices report some robustness checks and a more detailed description of some of the variables.

2. Previous literature

Beginning with the seminal contribution of Albrecht et al. (2003), there has been a growing interest in how the gender wage gap varies throughout the wage distribution. In contrast with most of the previous studies that compare gender differences in average wages, this approach informs us about places in the wage distribution where these wage gaps appear and are more pronounced.[2] This recent literature generally finds larger gender wage gaps at the top of the wage distribution than in lower parts of the wage distribution, even after controlling for observable person, job and firm characteristics (Gardeazabal and Ugidos 2005; Nordman and Wolff 2009; Datta Gupta et al. 2006; Arulampalam et al. 2007; De la Rica et al. 2008; Jellal et al. 2011). These results, all of which were obtained using data from European labour markets, are consistent with what is referred to as a *glass ceiling effect*. It is of substantial interest to learn whether wage patterns are similar in Canada, where government policies, wage-setting institutions, women's relative market qualifications and the wage structure differ in important ways from European environments.

The economy-wide glass ceiling effect found by these studies could be driven by two different mechanisms, one operating within firms and the other between firms. It could stem from the fact that females are sorted into lower-paying jobs within firms compared to their male counterparts (i.e., they face within-firm glass ceilings). However, as suggested by Dickens and Katz (1987), Groshen (1990 and 1991a), Bronars and Famulari (1997), Salvanes et al. (1999), Abowd et al. (1999) and Woodcock (2007), inter-firm wage differentials explain a large portion of variation in individuals' wages. Therefore, the second mechanism that could give rise to the economy-wide glass ceiling effect faced by females is the disproportionate sorting of men and women across high-paying and low-paying firms (the glass door effect). Similar to the glass ceiling effect that leads to under-

2 For instance, any observed gender differences in average wages could be driven by large wage gaps faced by females at the top of the wage distribution or at the bottom of the wage distribution, or by a uniform wage gap faced by females throughout the wage distribution. These different wage patterns have different implications and require different remedies.

representation of females in high-wage regions of the wage distribution, the glass door effect leads to under-representation of females in high-paying firms. No previous studies of the glass ceiling faced by females distinguish between within-firm and economy-wide wage outcomes, nor do they contrast the extent to which the glass ceiling faced by females operates between firms with the extent to which it operates within firms.

A separate literature focuses on inter-firm sex segregation and its effect on the average gender wage gap. Early studies of inter-firm sex segregation use specialized samples that are not representative of the national economy and cover only a narrow range of industries, occupations or regions. They find that women are typically segregated into lower-paying firms, even within occupations, and this inter-firm sex segregation accounts for a considerable portion of the average gender wage gap (McNulty 1967; Buckley 1971; Blau 1977; Pfeffer and Davis-Blake 1987; Groshen 1991b; Reilly and Wirjanto 1999; Carrington and Troske 1995, 1997, 1998). More recent studies (Meyersson Milgrom et al. 2001; Bayard et al. 2003; Datta Gupta and Rothstein 2005; Amuedo-Dorantes and De la Rica 2006) use nationally representative matched employer-employee data sets to quantify the contribution of gender segregation by industry, occupation, establishment and occupation-establishment cell (job cell) to the average gender wage gap. All these studies find that segregation of females into lower-paying occupations, industries, establishments and occupations within establishments accounts for a substantial portion of the average gender wage gap. With the exception of Meyersson Milgrom et al. (2001), they also find that there remains a considerable within-job-cell average gender wage gap even after controlling for observed worker characteristics and accounting for segregation of females into lower-paying occupations, industries, establishments and occupations within establishments.

Pendakur and Woodcock (2010) develop a methodology in the context of native-immigrant wage disparities which has several advantages over methodologies used in recent studies of inter-firm sex segregation (Bayard et al. 2003; Datta Gupta and Rothstein 2005; Amuedo-Dorantes and De la Rica 2006). First, it provides a way to measure the effect of inter-firm segregation throughout the wage distribution, not only at the mean. Second, it does not rely on estimates of the proportion of females in the firm to quantify the effects of inter-firm gender segregation on the gender wage gap.[3] As pointed out by Bayard et al. (2003) because only a sample of a firm's employees are observed in these data, sampling error in these estimates can be severe. In addition, using the estimates of the proportion of females in each firm to study the inter-firm gender segregation assumes a particular functional form on the way segregation affects wages.

3 For example, to quantify the effect of inter-firm gender segregation, Bayard et al. (2003) use a regression of wages on the estimates of the proportion of females in a worker's firm, as well as usual observable characteristics.

3. Empirical methodology

Estimating the gender wage gap at several quantiles of the conditional wage distribution enables me to assess the existence of an economy-wide glass ceiling experienced by females. The existence of an economy-wide glass ceiling would imply that females are under-represented (over-represented) in high- wage (low-wage) regions of the wage distribution, and that their under-representation becomes more pronounced as we move up the wage distribution. As a consequence, in the presence of an economy-wide glass ceiling, the gap between male and female wages will be larger at the top of the wage distribution than at the middle or bottom.

I measure the wage gap at the τ-th conditional quantile of the wage distribution using the following quantile regression: [4]

$$\mathrm{q}_\tau(w_{ij}|\mathrm{x_{ij}}, \mathrm{g_{ij}}) = x'_{ij}\beta^\tau + g_{ij}\delta^\tau, \tag{1}$$

where w_{ij} is the log hourly wage of worker i at firm j; x_{ij} is a vector of observable characteristics that influence wages (e.g., education, labour market experience and ethnicity); g_{ij} is a gender indicator which is equal to one for females, and $\mathrm{q}_\tau(w_{ij}|\mathrm{x_{ij}}, \mathrm{g_{ij}})$ is the τ-th conditional quantile of w_{ij}.[5] Therefore, β^τ measures the marginal change in the τ-th conditional quantile of the wage distribution due to marginal change in an observed characteristic, and δ^τ measures the difference between the τ-th quantile of log wages of males and females, conditional on X_{ij}.[6] Comparing estimates of δ^τ at different quantiles enables me to study how gender wage differentials vary over the conditional wage distribution and allows me to examine the existence of a glass ceiling faced by females.

3.1. Glass doors

The glass door effect arises if women are disproportionately sorted into lower-paying firms compared to their male counterparts. Such sorting will contribute to the economy-wide gender wage gap. I apply a methodology developed by Pendakur and Woodcock (2010) in the context of native-immigrant wage differentials to measure the glass door effect by comparing within-firm and economy-wide gender wage gaps. I measure the effect of glass doors on the economy-wide conditional average gender wage gap to examine the extent to which it is driven by females' sorting into lower-paying firms compared to observationally equivalent males. I also measure the glass door effect at different quantiles of the conditional wage distribution to quantify the extent to which the gender wage gap in different parts of the wage distribution and its pattern of change throughout the wage

4 Alternatively, the model that I am seeking to estimate can be expressed as a quantile regression that satisfies $\Pr[W'_{ij}\beta^\tau + g_{ij}\delta^\tau | X_{ij}, g_{ij}] = \tau$.

5 All the regressions in this paper also include year effects that are suppressed from regression equations for expositional simplicity.

6 As it is discussed in section 4, I also estimate equation (1) for different subsamples to allow for heterogeneity in observable dimensions.

distribution is driven by gender differences in sorting across firms. This enables me to measure the contribution of glass doors to the economy-wide glass ceiling.

To quantify the contribution of glass doors to the economy-wide conditional average gender wage gap I need to construct economy-wide and within-firm measures of average gender wage gap. I use the following linear regression model to estimate the economy-wide average gender wage gap, conditional on observed individual and job characteristics:

$$w_{ij} = x'_{ij}\beta + g_{ij}\delta + \varepsilon_{ij}, \tag{2}$$

where ε_{ij} is a stochastic mean-zero error term. The estimates of the within-firm average gender wage gap could be obtained by adding firm effects to equation (2). In the mean regression case, we have:

$$w_{ij} = x'_{ij}\beta + g_{ij}\delta^{*} + f'_{ij}\psi + \varepsilon_{ij}, \tag{3}$$

where f_{ij} is a vector of indicators for each firm that is equal to 1 if worker i is employed at firm j, and ψ is a vector of firm effects that measure inter-firm differences in average wages, conditional on worker and job characteristics X_i and gender g_i. In equation (2), δ measures the average gender wage gap conditional on observed individual characteristics without making a distinction about workers' firm affiliation. Therefore, it does not take into account potential differences in the way men and women are sorted across high- and low-paying firms. In other words, the contribution of gender differences in sorting across firms to gender wage gap is part of δ in equation (2). In contrast, since δ^* in equation (3) is identified within firms, it measures the average gender wage gap isolating the contribution of gender differences in sorting across firms. Therefore, the contribution of gender differences in sorting across firms to gender wage gap is not part of δ^* in equation (3).

As a result, if the within-firm wage gap faced by females is smaller than the economy-wide wage gap, it implies that their low wage outcomes, relative to their male counterparts, are due partly to segregation into lower-paying firms.[7] As a result, $(\hat{\delta} - \tilde{\delta})$, where $\hat{\delta}$ and $\tilde{\delta}$ are estimates of δ and δ^* from equations (2) and (3), respectively, provides an estimate of the glass door effect. It measures how wages are affected by gender differences in sorting across firms, conditional on workers' observed characteristics.[8]

Under the null hypothesis of no gender differences in sorting across firms (i.e., a zero glass door effect), the inclusion of firm effects in the model will not affect

7 Petersen et al. (2011) use the same approach to measure the contribution of sorting across high-paying and low-paying firms, occupations, and occupation-firm units to average wage gap between married and single men. Pendakur and Woodcock (2010) provide a formal proof for this approach and apply it to measure the effect of sorting across firms on native-immigrant wage differentials. Gelback (2009) also proposes a conditional decomposition method for mean regressions similar to Pendakur and Woodcock (2010).

8 It should be emphasized that the estimated glass door effect does not necessarily measure discrimination that leads to inter-firm sex segregation. It could also measure the effect of unobservable factors that are correlated with gender and firm affiliation (such as physical strength, preferences or productivity). These issues will be discussed in section 4.2.

the consistency of the estimate of the gender wage gap (i.e., both equations (2) and (3) provide consistent estimates of the gender wage gap). However, estimated gender wage gap in the specification that includes firm effects will be inefficient due to inclusion of irrelevant variables (i.e., firm indicators). Under the alternative hypothesis of the existence of gender differences in sorting across firms (i.e., a non-zero glass door effect), however, only the estimate of the gender wage gap from the specification that includes firm effects is consistent. This motivates a Hausman test for the presence of a glass door effect:[9]

$$H = \frac{(\hat{\delta} - \tilde{\delta})^2}{var[\tilde{\delta}] - var[\hat{\delta}]} \sim \chi_1^2. \tag{4}$$

I measure the contribution of glass doors to the gender wage gap at different points of the wage distribution in an analogous fashion. The quantile regression model with firm effects may be expressed as:

$$\mathrm{q}_\tau(w_{ij}|\mathrm{x_{ij}}, \mathrm{g_{ij}}, f_{ij}) = X_{ij}'\beta^\tau + g_{ij}\delta_*^\tau + f_{ij}'\psi. \tag{5}$$

Using a two-stage procedure proposed by Canay (2011), I first use an OLS regression to estimate the average firm effects conditional on worker and job characteristics. In the second stage, I subtract these estimates from my dependent variable and proceed using the standard quantile regression method to regress this transformed variable on my gender indicator and other observed characteristics. Canay (2011) shows that under the assumption that firm effects do not change with different quantiles, this procedure removes the fixed effects and gives consistent estimates of the slope parameters.

Pendakur and Woodcock (2010) show that if the wage determination model (equation (5) in my case) is correctly specified, then $(\hat{\delta}^\tau - \tilde{\delta}^\tau)$ has a similar interpretation as the mean regression case, where $\hat{\delta}^\tau$ and $\tilde{\delta}^\tau$ are coefficients on g_i in quantile regressions that include (equation (5)) and exclude (equation (1)) firm effects, respectively. Specifically, $(\hat{\delta}^\tau - \tilde{\delta}^\tau)$ provides a measure of the glass door effect at the τ-th quantile of the conditional wage distribution. As in the mean regression case, I can also test for the glass door effect at a particular quantile using a Hausman test.

3.2. Data and sample characteristics

My estimates are based on the Workplace and Employee Survey (WES), which is a Canadian linked employer-employee data set. The survey was administered from 1999 to 2005. The employer sample is longitudinal and refreshed every second year (i.e., in 2001, 2003 and 2005) to maintain a representative cross-section. The target population of employers is all business locations in Canada that have paid employees in March of each surveyed year, except employers in Yukon, Nunavut and the Northwest Territories and employers operating in crop production and

9 See Pendakur and Woodcock (2010) for the proof.

animal production; fishing, hunting and trapping; private households; religious organizations; and public administration.[10]

A maximum of 24 employees were sampled from each firm in each odd year and were followed the next year.[11] My analysis is based on the pooled 1999, 2001, 2003 and 2005 cross-sections. The data from even-numbered years are not used in order to avoid sample selection problems associated with employee attrition in these years.

I restrict the sample to non-Aboriginal workers between the ages of 24 and 65. I also restrict the sample of firms to those that report on average six or more employees per year, and that have at least two workers sampled over the entire period they appear in the data. The restricted sample includes 30,115 women and 40,230 men from 6,584 firms. The number of employees observed in each firm, pooled across all years a firm appears in the data, ranges from 2 to 63, with an average number of 16 and a median of 15. I observe 2,373 firms in all four years, 1,519 firms in three years, 1,262 firms in two years and the remaining 1,430 in one year.

I estimate wage differentials for the sample of all workers, as well as for seven different subgroups including females with at least one dependent child and all males, females without dependent children and all males, single workers without dependent children, workers younger than 44 years of age, workers older than 43 years of age, workers with a bachelor's degree or higher and workers without a bachelor's degree.[12]

My dependant variable is the natural logarithm of hourly wages.[13] The individual characteristics used in my main regression specifications (reported in table 2) are: highest level of schooling (8 categories), marital status (6 categories), age

10 Public administration comprises establishments engaged primarily in activities of a governmental nature, that is, the enactment and judicial interpretation of laws and their pursuant regulations, and the administration of programs based on them. Legislative activities, taxation, national defence, public order and safety, immigration services, foreign affairs and international assistance, and the administration of government programs are activities that are purely governmental in nature (see Industry Canada's Canadian Industry Statistics, available at ic.gc.ca/cis-sic/cis-sic.nsf/IDE/cis-sic91defe.html). Public administration's share of employment in Canada is around 6.5% (Statistics Canada, Table 281-0024, available at www5.statcan.gc.ca/cansim/a26?lang=eng&id=2810024. Accessed October 15, 2011)).

11 The number of workers sampled from each firm was proportional to the firm's size except workplaces with fewer than four employees where all employees were selected.

12 I use the sample of all workers and appropriate interactions with gender to identify the wage differentials for these subgroups. For instance, to estimate the average wage gap for workers younger than 44 and older than 43, I use the following regression: $W_{ij} = X_{ij}'\beta + FEMALE * YOUNGER44 + FEMALE * OLDER43 + YOUNGER44$, where "YOUNGER44" and "OLDER43" are indicators for workers younger than 44 and older than 43, respectively.

13 The hourly wage measure that I use in my analysis is constructed by Statistics Canada and is directly available in the WES. There are several ways for workers to report their wages or salaries in the survey, and Statistics Canada uses this information to convert the reported wages/salaries to hourly measures using other information gathered from the employees such as usual hours of work per week (measured separately for those with regular and irregular weekly hours) and number of weeks and months worked per year (including paid leave). The hourly wage measure includes extra earnings such as overtime, bonus, profit sharing, etc.

(9 categories), number of dependent children (5 categories), a quartic in years of (actual) full-time labour market experience, a quadratic in years of seniority with the current employer, an indicator for full-time employment, an indicator for membership in a union or collective bargaining agreement, and indicators for being a Canadian-born visible minority, white immigrant or a visible minority immigrant.

All regression models are estimated using employee sample weights provided by Statistics Canada. Following Statistics Canada's suggested procedure, I use 100 sets of bootstrap sample weights provided by Statistics Canada to estimate standard errors in my regressions.[14]

Table 1 reports weighted sample means for males and females. In comparison to males, the average female is more educated, less likely to have children and has fewer years of full-time labour market experience. In terms of job characteristics, the average female has fewer years of employer seniority, is less likely to work full time, more likely to be a member of a union or collective bargaining agreement, less likely to work flexible hours, less likely to be able to carry out work duties at home and less likely to be a manager. In terms of employer characteristics, the average female is more likely to work for a non-profit enterprise and more likely to work for an employer with an employment or pay equity program.

4. Results

4.1. Glass ceilings and glass doors

Figure 1 illustrates the distribution of the proportion of female employees at firms in which male and female workers are employed.[15] As the figure suggests, firms tend to be quite segregated by gender. Males on average are employed at firms where only 32% of workers are female, while females are on average employed at firms where 62% of workers are female. Figure 2 illustrates the distribution of estimated firm effects from equation (3) for four different categories of firms: firms with a proportion of female employees below 25%, between 25% and 50%, between 50% and 75% and above 75%. Firm effects are on average larger for firms that employ fewer females, even after controlling for differences in workers' personal and job characteristics across firms, including gender.[16] Together, figures 1 and 2 suggest that there is significant gender segregation at the firm level, and firms that employ fewer females tend to pay higher wages, conditional on observed individual and job characteristics. This result highlights the importance of studying the effect of gender segregation across firms on gender wage gap.

14 The bootstrap weights will account for the potential non-independence of error terms for workers within the same firm. They will also correctly adjust for the variation due to the two-stage sampling of employees, as well as the complex survey design of the WES (Drolet 2002).

15 Both figure 1 and 2 are generated using data from 2003 and 2005. This is because the total number of female workers in each firm is a new variable added to WES in 2003.

16 In other words, workers (both males and females) employed at firms with a higher proportion of females receive lower wages on average than those employed at firms with a lower proportion of females.

TABLE 1
Sample means

	Males	**Females**
No. of observations	40,230	30,115
Subsamples		
Workers with at least one dependent child	52.53	50.12
Workers without dependent children	47.47	49.88
Workers younger than 44	43.84	43.95
Workers older than 43	56.16	56.05
Single workers without any child	18.20	18.54
Personal characteristics		
Hourly wage	24.12	19.01
Log hourly wage	3.11	2.88
Standard deviation of log hourly wage	0.47	0.45
Age	42.03	41.99
Years of (actual full-time) experience	20.01	16.70
Ethnicity		
Visible minority Canadians	0.015	0.017
Visible minority immigrants	0.092	0.090
White immigrants	0.116	0.104
White Canadians*	0.775	0.787
Highest educational attainment		
PhD, master's or MD	0.052	0.044
Other graduate degree	0.023	0.024
Bachelor's degree	0.143	0.147
Some university	0.081	0.085
Completed college	0.168	0.253
Some college or trade certificate	0.248	0.204
High school diploma	0.162	0.167
Less than high school*	0.119	0.072
Marital status		
Married	0.623	0.581
Common law	0.144	0.132
Separated	0.023	0.035
Divorced	0.035	0.077
Widowed	0.004	0.015
Single*	0.168	0.157
No. of dependent children		
0*	0.474	0.498
1	0.177	0.182
2	0.247	0.235
3	0.078	0.068
4+	0.022	0.014
Job characteristics		
Full-time	0.870	0.589
Member of union or collective bargaining agreement	0.288	0.306
Years of seniority with current employer	9.717	8.817
Possibility to work flexible hours†	0.375	0.322
Possibility to carry out work duties at home†	0.280	0.238

NOTES: * indicates reference category for regressions. † indicates that the variable is not used in the main regressions and is used only for robustness checks. All the means are computed using sample weights provided in the data (Statistics Canada does not allow the reporting of these means without using the weights). Sample means for other variables (occupation, industry, degree of competition, firm size, etc.) are in table A4 in the online technical appendix available at cje.economics.ca.

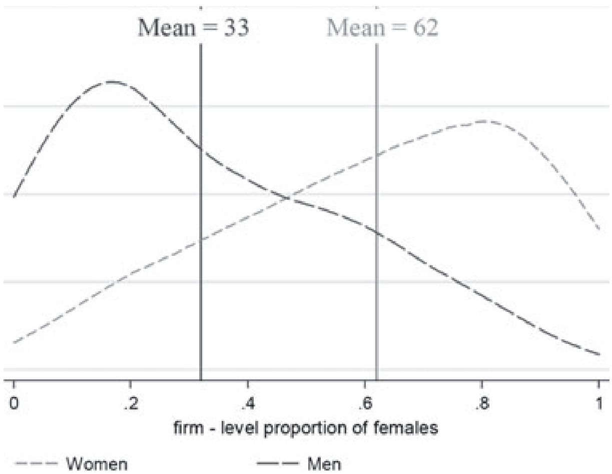

FIGURE 1 Distribution of firm-level proportion of females, for male and female employees

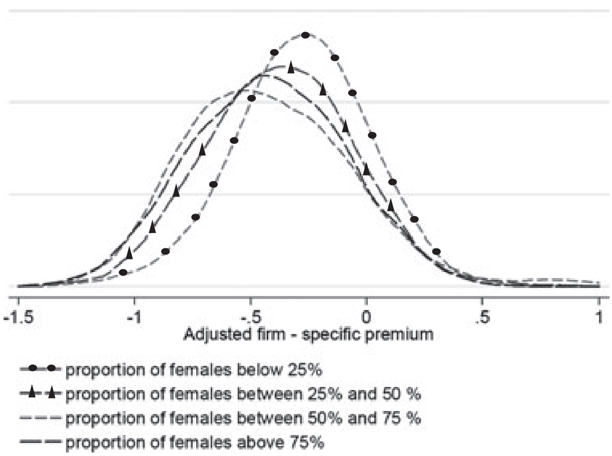

FIGURE 2 Distribution of firm effects by different firm-level proportion of females

One of the main objects of interest in this paper is to examine whether females face a glass ceiling, conditional on their observed characteristics. Looking at the unconditional gender wage gap, I find that the economy-wide gender wage gap is relatively flat throughout the wage distribution (−0.201 at the 10th percentile, −0.231 at the 50th percentile, −0.212 at the 90th percentile and −0.256 at the 95th percentile). However, this could be misleading as we might not be comparing quantile wages of comparable male and female workers (or in other words, we are not comparing the representation of comparable groups of male and female workers at different parts of the wage distribution). As reported in panel 1 of

TABLE 2
Gender wage gap and glass door estimates

	(1) All workers			(2) All single workers without any dependent child		
	Economy-wide	**Within firms**	**Glass door**	**Economy-wide**	**Within firms**	**Glass door**
Mean differential	−0.183***	−0.162***	−0.021***	−0.085***	−0.109***	0.024*
	(0.009)	(0.007)	[0.000]	(0.017)	(0.011)	[0.064]
Quantile differential						
10th percentile	−0.137***	0.136***	−0.001	−0.005	−0.105***	0.100***
	(0.012)	(0.008)	[0.974]	(0.037)	(0.021)	[0.000]
Median	−0.192***	0.168***	−0.024***	−0.077***	−0.132***	0.055***
	(0.009)	(0.006)	[0.000]	(0.021)	(0.014)	[0.000]
90th percentile	−0.235***	0.170***	−0.066***	−0.160***	−0.117***	−0.043***
	(0.024)	(0.019)	[0.000]	(0.038)	(0.020)	[0.000]
No. of observations	70,345	70,345		70,345	70,345	

	(3) All female workers without any dependent child and all male workers			(4) All female workers with at least one dependent child and all male workers		
	Economy-wide	**Within firms**	**Glass door**	**Economy-wide**	**Within firms**	**Glass door**
Mean differential	−0.176***	−0.168***	−0. 008	−0.195***	−0.159***	−0.036***
	(0.009)	(0.007)	[0.157]	(0.013)	(0.009)	[0.000]
Quantile differential						
10th percentile	−0.119***	−0.138***	0.019***	−0.157***	−0.136***	−0.021***
	(0.018)	(0.013)	[0.004]	(0.018)	(0.015)	[0.000]
Median	−0.182***	−0.176***	−0.006	−0.205***	−0.165***	−0.040***
	(0.013)	(0.008)	[0.421]	(0.012)	(0.008)	[0.000]
90th percentile	−0.236***	−0.182***	−0.054***	−0.240***	−0.159***	−0.081***
	(0.030)	(0.017)	[0.000]	(0.036)	(0.025)	[0.000]
No. of observations	70,345	70,345		70,345	70,345	

	(5) All workers younger than 44			(6) All workers older than 43		
	Economy-wide	**Within firms**	**Glass door**	**Economy-wide**	**Within firms**	**Glass door**
Mean differential	−0.191***	−0.152***	−0.039***	−0.171***	−0.175***	0.004***
	(0.012)	(0.007)	[0.000]	(0.011)	(0.011)	[0.000]
Quantile differential						
10th percentile	−0.150***	−0.135***	−0.015	−0.115***	−0.141***	0.026
	(0.018)	(0.013)	[0.228]	(0.023)	(0.016)	[0.527]
Median	−0.209***	−0.166***	−0.043***	−0.163***	−0.175***	0.012
	(0.0138)	(0.007)	[0.000]	(0.020)	(0.009)	[0.102]
90th percentile	−0.254***	−0.161***	−0.093***	−0.256***	−0.184***	−0.072***
	(0.034)	(0.024)	[0.000]	(0.028)	(0.021)	[0.000]

TABLE 2
(Continued)

	(7) All workers with bachelor's degree or higher			**(8) All workers without bachelor's degree**		
	Economy-wide	**Within firms**	**Glass door**	**Economy-wide**	**Within firms**	**Glass door**
Mean differential	−0.150***	−0.119***	−0.031	−0.193***	−0.177***	−0.044***
	(0.026)	(0.012)	[0.179]	(0.008)	(0.008)	[0.000]
Quantile differential						
10th percentile	−0.123***	−0.090***	−0.033	−0.140***	−0.145***	0.005
	(0.031)	(0.023)	[0.112]	(0.014)	(0.010)	[0.610]
Median	−0.139***	−0.127***	−0.012	−0.208***	−0.181***	−0.027***
	(0.022)	(0.016)	[0.416]	(0.011)	(0.005)	[0.005]
90th percentile	−0.213***	−0.143***	−0.070***	−0.247***	−0.176***	−0.071***
	(0.077)	(0.053)	[0.000]	(0.017)	(0.015)	[0.000]
No. of observations	70,345	70,345		70,345	70,345	

NOTES: Standard errors are in parentheses, p-values for glass door test are in brackets. Simulated standard errors for estimates of females' representation are all less than 0.002. Given the precision of our estimates, we omit standard errors from the tables to minimize clutter. Details are available on request. *** indicates statistically significant at 1% and * indicates statistically significant at 10%. All coefficients are estimated using sampling weights provided by Statistics Canada and all the standard errors are computed using 100 sets of bootstrap weights provided by Statistics Canada. All regressions are based on pooled samples of all males and females. Gender wage gap estimates for different subsamples are generated using interaction between gender and appropriate indicators.

table 2 and discussed below, once I start accounting for differences in observed characteristics between men and women, I find clear evidence of a glass ceiling effect faced by females that is robust across different subsamples. This suggests that differences in characteristics explain a bigger part of the gender wage gap at lower parts of the wage distribution than at the top, while the contribution to the gender wage gap of gender differences in returns to these characteristics increases as we move up the wage distribution, giving rise to an increasing conditional gender wage gap. This is also consistent with the findings of Albrecht et al. (2003) and Firpo et al. (2011). Albrecht et al. (2003) find that "it is clearly not gender differences in age, education, and immigration status that account for the gender wage gap at the top of the distribution but, rather, the differential rewards that women receive for these characteristics." Firpo et al. (2011) also find that the unconditional gender wage gap does not increase much at the top half of wage distribution (.249 at the 50th percentile and 0.258 at the 90th percentile). However, their decomposition results suggest that "gender differences in characteristics are much more important at the bottom (10th centile) than at the top (90th centile) of the wage distribution. Indeed, some significant wage structure effects emerge at the 90th percentile."

Table 2 presents mean and quantile estimates of economy-wide and within-firm gender wage gaps, and estimates of the glass door effect, conditional on workers' characteristics. Results reported in panel 1 of table 2 reveal that females face substantial economy-wide and within-firm average wage gaps compared to their male counterparts: about -0.18 log points and −0.16 log points, respectively, in the sample of all workers. Almost 11% (−0.02 log points) of the economy-wide average wage gap that females face is due to the glass door effect. There is also strong evidence that females face an economy-wide glass ceiling: the wage gap increases from −0.13 log points at the 10th percentile of the conditional wage distribution to −0.19 log points at 50th percentile and −0.23 log points at the 90th percentile.[17]

Females also face within-firm glass ceilings, but their effects are much less pronounced than the economy-wide glass ceiling (−0.13 log points at the 10th percentile versus −0.17 log points at the 90th percentile). Therefore, most of the economy-wide glass ceiling, particularly at the top half of the wage distribution, is driven by inter-firm gender segregation (i.e., glass door effect). For example, out of a 10 percentage point increase in the economy-wide wage gap experienced by females between the 10th and 90th percentiles of the conditional wage distribution, which is evidence of the existence of a glass ceiling, only one third is due to the increase in the within-firm gender wage gap. The remaining two thirds is due to the increasing contribution of gender differences in sorting across firms. The evidence is even stronger at the top half of the distribution. Almost the entire increase in the economy-wide gender wage gap between the 50th and 90th percentile is driven by the glass door effect.

Figure 3 further illustrates the importance of the glass door effect in generating the economy-wide glass ceiling. The vertical distance between the red/circled line (the estimated economy-wide gender wage gaps) and the blue/diamonded line (the estimated within-firm gender wage gaps) measures the glass door effect at different points of the conditional wage distribution, which is represented by the green/squared line. The upward slope of the economy-wide gender wage gap across different quantiles (i.e., the economy-wide glass ceiling) is due mostly to the increasing contribution of inter-firm gender segregation (i.e., the glass door effect). Altogether, these results suggest that, conditional on their observed personal and job characteristics, females tend to sort into lower-paying firms than their male counterparts, and this differential sorting explains a substantial part of the gender wage gap. Moreover, this differential sorting contributes more to the gender wage gap at the top of the conditional wage distribution and hence drives the economy-wide glass ceiling.

Other panels of table 2 assess the robustness of these results. To examine whether the sorting of females into lower-paying firms is potentially the result of supply decisions made by women due to family issues, I present estimates of the gender wage gap for female workers with and without dependent children compared to all male workers. Results reported in panels 2 and 3 of table 2 suggest that both groups face wage gaps similar to the overall sample of female workers, although economy-wide wage gaps are slightly larger for women with dependent

17 It might be worth mentioning that my estimates of the gender wage gap do not change much between the 90th and 95th percentile of the conditional wage distribution.

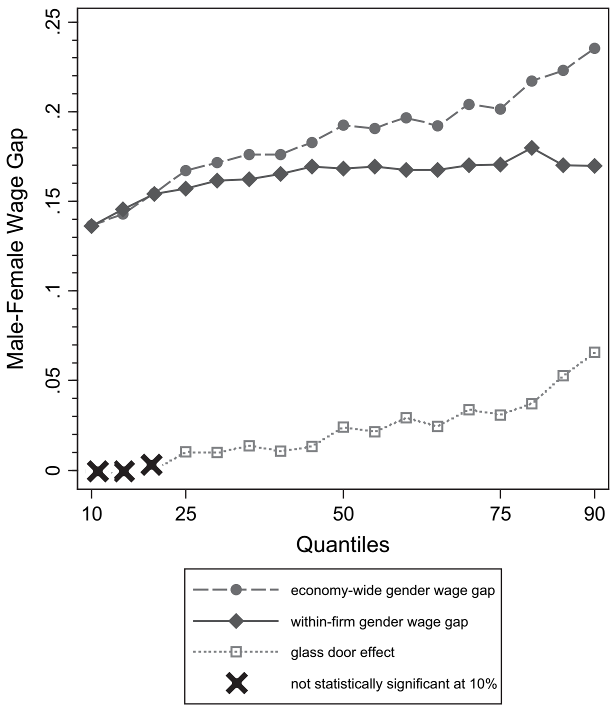

FIGURE 3 Sample of all workers – Conditional quantile gender wage gaps

children than women without dependent children. Both groups face a substantial economy-wide glass ceiling that is strongly driven by the glass door effect. Similar to the results reported for the sample of all workers, the glass door effect is almost entirely responsible for the glass ceiling faced by females at the top half of the wage distribution.

To further examine the role of family issues, panel 2 of table 2 presents estimates for all single workers without dependent children. In this group, there are fewer concerns regarding self-selection of females into lower-paying firms due to child or family responsibilities, which some might argue are partly responsible for the observed patterns. The estimates reveal that single female workers also experience a substantial, though smaller economy-wide mean wage gap (−0.085 log points). Interestingly, the average glass door effect is positive for this group, which implies single females without dependent children on average sort into higher-paying firms compared to their male counterparts. One might therefore conclude that the sole source of the average economy-wide wage gap that single females experience is the within-firm wage gap.

Estimated wage gaps at different quantiles of the conditional wage distribution, however, contradict such a conclusion. Similar to previous subgroups, single female workers experience a substantially significant economy-wide glass ceiling that is driven mainly by the glass doors. The economy-wide wage gap is small and statistically insignificant at the bottom decile of the conditional wage distribution, while there exists a significantly large within-firm gender wage gap (−0.10 log points). This large within-firm gender wage gap is offset by a positive glass door effect. This result suggests that at the bottom of the conditional wage distribution, single females tend to be employed at higher-paying firms than their male counterparts. At the 50th percentile of the conditional wage distribution, females face substantial economy-wide and within-firm wage gaps (−0.077 log points and −0.13 log points, respectively). Again, a positive glass door effect (0.055 log points) offsets part of the within-firm gender wage gap. At the top decile of the conditional wage distribution, however, the economy-wide gender wage gap is more than twice as large as the median economy-wide gender wage gap (−0.16 log points), which is attributed mainly to a large negative glass door effect (−0.043 log points). In fact, the contribution of the glass doors to the glass ceiling effect at the top half of the wage distribution (i.e., the increase in the economy-wide wage gap between the 50th and the 90th percentile) is larger for single females without dependent children compared to all females. Figure 8 further highlights the contribution of glass doors to the economy-wide glass ceiling faced by single females without children.

As the next robustness check, I present estimates of the gender wage gap for workers with different education levels. If gender differences in sorting across firms are supply-driven, then we would not expect to see the same wage gap patterns for highly educated females with potentially higher labour market attachments. Comparing these two groups also has another important implication. Comparing conditional quantiles of the gender wage gap for all workers tells us that females are under-represented in high-wage regions of the conditional wage distribution. It does not tell us directly whether this is due to under-representation of highly educated and skilled females at high-wage jobs in the economy (which is typically how people might interpret the glass ceiling effect), or due to under-representation of low-educated females at jobs that are considered high-paying for them but are not necessarily high-paying jobs in the economy. Both of these scenarios provide evidence for the existence of a glass ceiling effect, experienced, however, by two different groups of females and therefore driven by different mechanisms. It is therefore interesting and important to understand whether the estimated glass ceiling effect faced by all females is experienced by only one of these groups, or both.

Results reported in panels 7 and 8 of table 2 suggest that there are no significant differences between females with a bachelor's degree or higher and females without a bachelor's degree. There is strong evidence that both groups face a glass ceiling effect. The gender wage gap increases by 0.09 log points and 0.10 log points between the 10th and 90th percentile of the conditional wage distribution for females with a bachelor's degree or higher and females without a bachelor's degree, respectively. There is also clear evidence that the glass ceiling effect faced by both groups is driven mainly by glass doors, especially at the top half of the wage distribution.[18] The existence of a glass ceiling effect for females with a bachelor's degree or higher lines up more closely with the popular notion of a glass ceiling effect because it is based on gender wage gaps among highly educated and skilled individuals who are competing for high-wage jobs in the economy. The existence of a glass ceiling effect for females without a bachelor's degree further suggests that the effect also exists for those relatively low-educated females competing for jobs that are considered high-paying for them, given their characteristics, but are not necessarily high-paying jobs in the economy.

As the last robustness check, I estimate the gender wage gaps for younger and older workers. As Albrecht et al. (2003) point out, a potential explanation for an economy-wide glass ceiling is a compositional effect. If females' labour market prospects have improved over time relative to males, then the wage gap between older men and women will be larger than the gap between younger men and women. On the other hand, since wages increase with experience in the labour market, older workers tend to have higher wages than younger workers. The combination of these two factors could generate an increasing gender wage gap as we move up the wage distribution, which looks like a glass ceiling. The estimates reported in panels 5 and 6 of table 2 rule out this hypothesis. Here we observe that younger females experience larger economy-wide wage gaps than older females compared to their male counterparts. Moreover, both groups experience a substantial economy-wide glass ceiling that is driven mainly by the glass door effect.

Altogether, results reported in panels 1 through 8 of table 2 and figures 3 to 10 provide strong evidence that females experience a glass ceiling effect that is driven mainly by gender differences in sorting across firms, especially at the top half of the wage distribution. These results hold for different subgroups of workers and cannot be explained by females' supply decisions driven by family issues, education level (which could be a proxy for the type of jobs females are competing for, with high-education females competing for higher-paying jobs, or could be a proxy for the degree of labour market attachment) or changes in labour market prospects for younger workers. Online appendix A also presents estimates of a number of additional robustness checks. In each case, these alternative specifications yield estimates similar to those reported in the main text.[19]

18 Estimates of the gender wage gap for workers with a college degree or higher and workers without a college degree also exhibit the same patterns. These results are not reported here due to their similarity to the results reported in panels 7 and 8 but are available upon request.

19 These tables are available in a technical appendix available at cje.economics.ca. Table A1 re-estimates the above regressions by adding some additional control variables including family income from employment (excluding worker); family income from other sources; four indicators for people who are willing but unable to work more hours due to unavailability of childcare, family responsibilities, going to school and transportation problems; an indicator for possibility to work flexible hours; and an indicator for possibility to carry out work duties at home. Table A2 excludes the immigrant workers from the sample. Table A3 allows the returns to observable characteristics to differ across different subgroups. Table A4 includes occupation (6 categories) as a control.

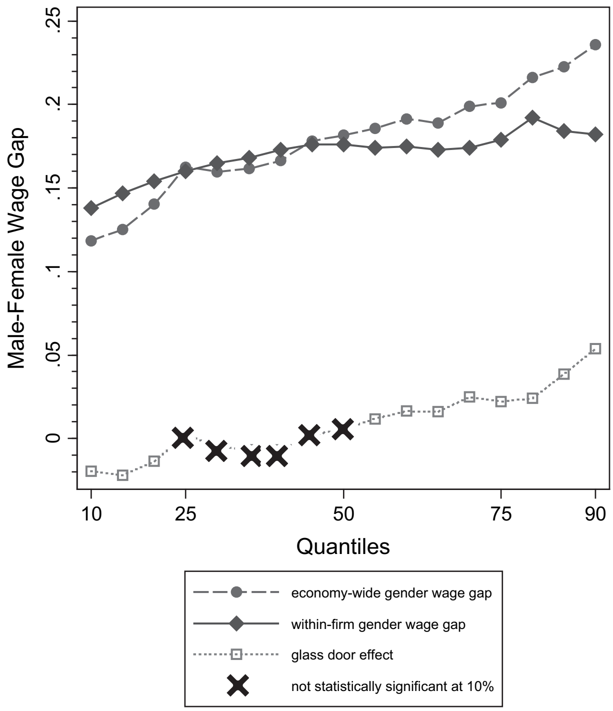

FIGURE 4 All female workers without any dependent child and all male workers – Conditional quantile gender wage gaps

4.2. Possible explanations: Compensating differentials versus efficiency wages

My measure of glass ceiling is based on gender differences in wages that are not explained by observed individual and job characteristics that could be correlated with gender and also affect wages. However, one could argue that the residual gender differences in wages across quantiles and the glass ceiling effect may reflect gender differences in unobserved characteristics such as productivity or preferences, not discrimination. In the light of this argument, this section investigates the potential explanations for the inter-firm gender segregation that gives rise to the glass ceiling faced by females.

If unobserved gender differences in productivity are driving the glass ceiling effect, this would imply that men are more productive than women, and this productivity gap becomes larger as we move up the wage distribution. As various studies suggest, women have smaller probabilities of promotion into high-paying jobs, and therefore reaching high-wage regions of the wage distribution, compared to their male counterparts. Regardless of the underlying reasons behind gender differences in promotion opportunities (discrimination, unmeasured differences in productivity, commitment, etc.), the sequential selection effect

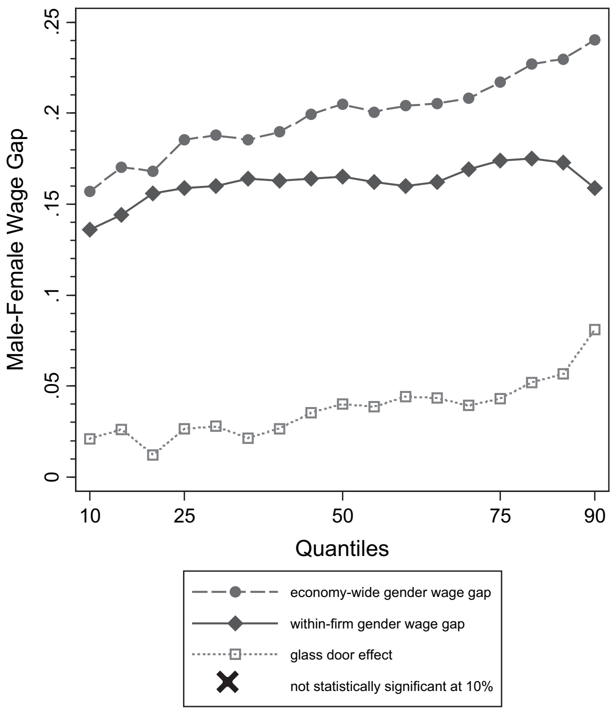

FIGURE 5 All female workers with at least one dependent child and all male workers – Conditional quantile gender wage gaps

generated through this process should reduce the unobservable differences between men and women by the time they make it to the top of wage distribution.

For example, imagine that we have 120 male and 120 female workers, and there are two levels of management hierarchy. Women have two thirds the chance of being promoted. Therefore, 80 women and 120 men will be promoted to the second level. If females' lower chance of promotion was due purely to discrimination, since these 80 females were more stringently selected compared to their male colleagues, they will have better job-related characteristics than their male counterparts, and therefore should face lower wage gaps. If the lower odds of promotion were due purely to differences in unobserved factors such as productivity, these differences should be on average smaller between men and women in the second level, and again women should face lower wage gaps. Therefore, if anything, we would expect these unobservable differences to have smaller effects on residual gender wage gaps as we move up the wage distribution, which contrasts with the patterns I find.

The theory of compensating differentials could provide an alternative explanation for inter-firm wage differentials and subsequently gender differences in sorting across firms, which could be used to explain the glass ceiling effect. In

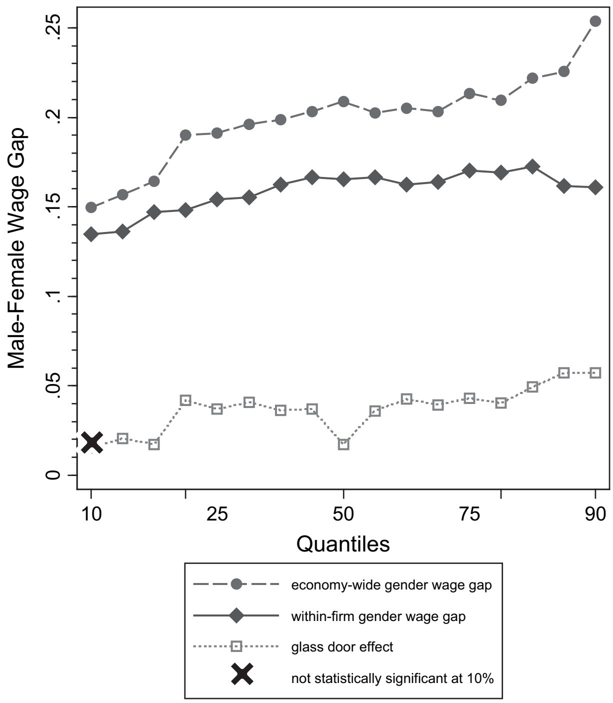

FIGURE 6 All workers younger than 44 – Conditional quantile gender wage gaps

the context of a compensating differentials model, the inter-firm wage differentials stem from inter-firm differences in working conditions. For example, high-paying firms offer higher wages on average but also have relatively less flexible or less pleasant working conditions. If men and women value job characteristics differently—because of family issues or gender differences in preferences, for instance—then females will choose low-paying firms because they care more about non-pecuniary aspects of a job, while men will choose high-paying firms because they care more about wages. The gender pay differentials may be therefore compensated by other characteristics of female jobs such as less demanding or more flexible working conditions.[20] As a result, if inter-firm differences in working conditions get larger as we move up the wage distribution, this could generate an increasing gender wage gap that looks like a glass ceiling.

20 The theory of compensating differentials is usually applied to explain the inter-occupational gender segregation and the resulting wage differentials. My results still hold when I control for different occupations and look at the segregation of females within occupations but across different firms (table A4, technical appendix available at cje.economics.ca). Filer (1985) and Jacob and Steinberg (1990) show that there are no significant differences in average measures of working conditions within occupations (even very broad categorization), and once we control for occupation, the effect of these measures is no longer significant. In addition, most of the studies that look at inter-industry wage differentials find no evidence in support of compensating differentials (Smith 1979; Brown 1980; Krueger and Summers 1988).

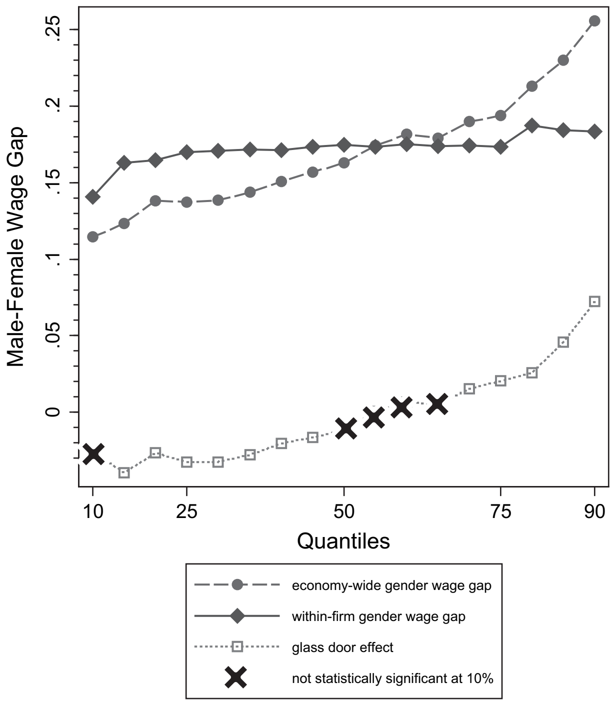

FIGURE 7 All workers older than 43 – Conditional quantile gender wage gap

As discussed before, results reported in table 2 for different subgroups suggest that supply-driven factors, such as family issues, cannot account for inter-firm gender segregation. To further investigate this issue, I estimate a specification that controls for job characteristics, including flexible work hours; the possibility to carry out work duties at home; and indicators of inability to work more hours due to unavailability of childcare, personal and family responsibilities, going to school or transportation problems. I also control for total family income from employment and other sources. These results are reported in the online technical appendix table A1 (available at cje.economics.ca) and are quantitatively and qualitatively similar to the main results in table 2. Aligned with previous results, this result also suggests that job characteristics or family issues do not explain the observed glass ceiling and glass door effects.

As a second test, I examine whether reported job satisfaction differs on average between similar workers with the same level of pay satisfaction employed in high-paying and low-paying firms. If wages compensate for undesirable job characteristics, and therefore higher average wages paid by high-paying firms are due to more undesirable working conditions at these firms, then workers employed at low-paying firms should report higher levels of job satisfaction, conditional on

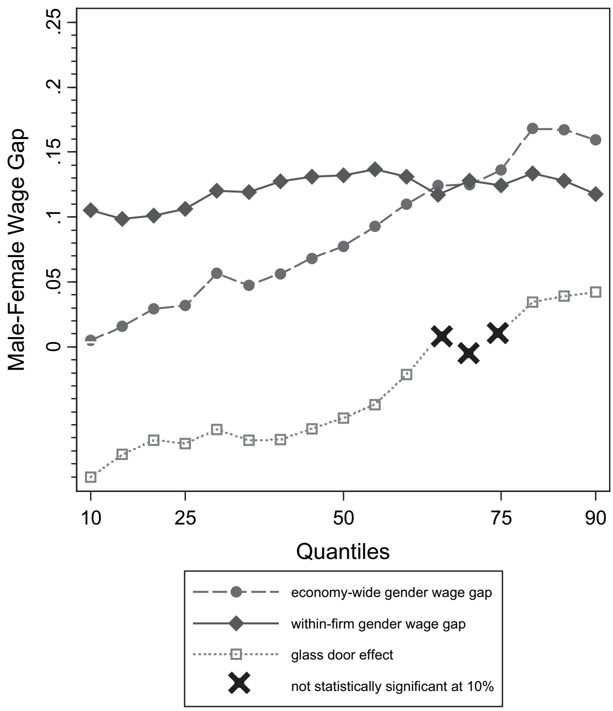

FIGURE 8 All single workers without dependent children – Conditional quantile gender wage gaps

workers' pay satisfaction (and other observed characteristics).[21] I run a regression of estimated firm effects from equation (3) on the observable worker and job characteristics used in my main specification, as well as four indictors for pay satisfaction and two indicators for job satisfaction. I run this regression for the sample of all workers and for male and female workers separately. The results are reported in table 3. Those employed at lower-paying firms do not report higher levels of job satisfaction, on average, than those employed at relatively higher-paying firms,

21 Obviously pay satisfaction (or wage) is part of a worker's notion of job satisfaction. Therefore, I am assuming that reported job satisfaction is the combination of the worker's satisfaction with working conditions and her pay satisfaction. Therefore, controlling for pay satisfaction is quite important here. Without controlling for pay satisfaction, we don't expect to see any difference in reported job satisfaction between workers employed at higher- versus lower-paying firms. Lower average satisfaction from working conditions at higher-paying firms is compensated by higher average pay satisfaction, while the opposite is true for workers at lower-paying firms. Therefore, the average job satisfaction, which is the combination of the two measures, would be on average the same across both groups. Controlling for pay satisfaction, however, enables me to compare workers with the same level of pay satisfaction (wages) and examine whether those employed at higher-paying firms, on average, report lower job satisfaction (due to harder working conditions), which is what the theory of compensating differentials would predict. I should mention that the reason I control for pay satisfaction rather than the worker's wage is that wages will be mechanically correlated with the dependent variable.

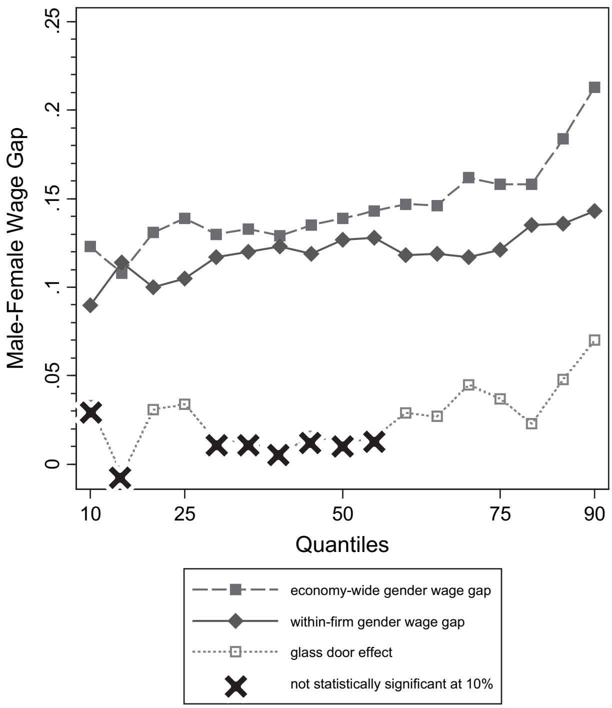

FIGURE 9 All workers with a bachelor's degree or higher – Conditional quantile gender wage gaps

conditional on observed characteristics and pay satisfaction.[22] Altogether, my results do not provide any evidence that compensating differentials contribute to the economy-wide glass ceiling through inter-firm gender differences in sorting.

Efficiency wage theory provides another potential explanation for inter-firm wage differentials. The efficiency wage hypothesis (See Shapiro and Stiglitz 1984, Stiglitz 1986, Bowles 1985, Bulow and Summers 1986, Yellen 1984 and Katz 1986) suggests that firms might find it profitable to pay above market clearing wages to increase effort, reduce shirking, lower turnover, attract a higher-quality workforce, increase productivity and improve worker morale and group work norms. At the same time, differences across firms in conditions dictating efficiency wages will lead to different optimal wages across firms. This implies that workers with identical productive characteristics might receive different wages depending on the firm at which they find employment. These different efficiency wages paid to workers with similar characteristics would not require compensating differentials if they are based on certain firm characteristics that do not have direct impacts on worker utility. Krueger and Summers (1988), Katz and Summers (1989) and Groshen (1991a) find empirical support in favour of efficiency wage theory.

22 These results are consistent for the sample of all workers, and for the samples of male and female workers.

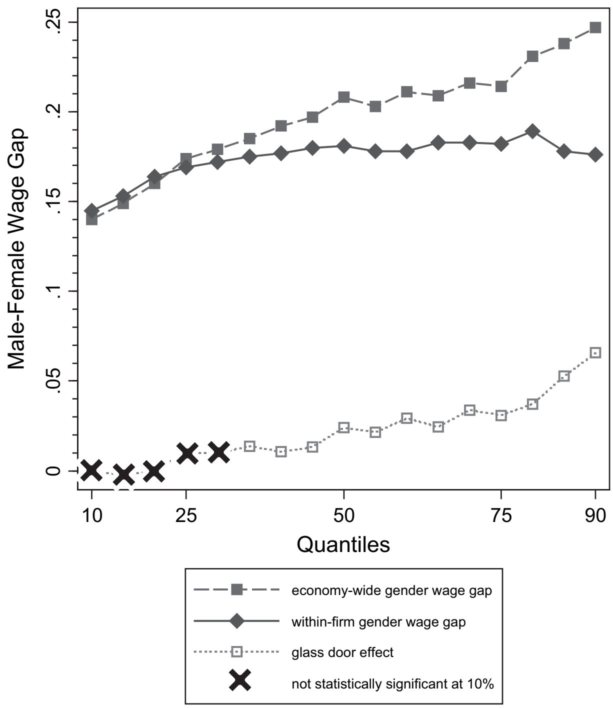

FIGURE 10 All workers without a bachelor's degree – Conditional quantile gender wage gaps

I test the predictions of efficiency wage theory regarding the relationship between firm-specific premiums and firm characteristics such as productivity and quit rates by regressing estimated firm effects from equation (3) on a wide set of firm-level characteristics.[23] The results are reported in table 4. Firms that pay higher premiums to their employees (after accounting for inter-firm differences in workforce composition) are on average larger, more likely to have a pay equity program, face more competition, are more likely to provide non-wage benefits, have lower quit rates, have higher training expenditures, have higher productivity, are more likely to have incentive compensation schemes and are less likely to have innovative work practices. These results support the predictions of the efficiency wage theory such as lower quit rates and higher productivity for higher-paying firms.

The question that still remains is whether efficiency wage theory can help us to understand the glass ceiling effect faced by females arising from inter-firm

23 This regression is only run for the sample of for-profit firms since the productivity measure is only available for these firms. Regressions run for all firms excluding productivity measures produce similar results. I also use other alternative measures of training and provision of different benefits by the firm, and I find the same qualitative results. These results are available upon request. Refer to the technical appendix at cje.economics.ca for a more detailed description of the variables.

TABLE 3
OLS regression results, firm-specific premiums and worker characteristics

	All workers	**Male only**	**Female only**
Female	−0.044***		
	(0.005)		
PhD, master's or MD	0.186***	0.166***	0.209***
	(0.013)	(0.018)	(0.018)
Other graduate degree	0.183***	0.185***	0.190***
	(0.011)	(0.019)	(0.018)
Bachelor's degree	0.183***	0.174***	0.198***
	(0.011)	(0.015)	(0.014)
Some university	0.112***	0.075***	0.153***
	(0.014)	(0.019)	(0.016)
Completed college	0.129***	0.115***	0.151***
	(0.009)	(0.012)	(0.013)
Some college or trade certificate	0.094***	0.078***	0.112***
	(0.008)	(0.011)	(0.012)
High school diploma	0.053***	0.031**	0.079***
	(0.008)	(0.012)	(0.012)
Married	0.039***	0.066***	0.009
	(0.007)	(0.010)	(0.009)
Common law	0.009	0.038***	−0. 022*
	(0.009)	(0.013)	(0.011)
Separated	0.020	0.037	0.002
	(0.017)	(0.027)	(0.020)
Divorced	0.039***	0.052***	0.016
	(0.010)	(0.014)	(0.012)
Widowed	0.008	0.032	−0. 016
	(0.021)	(0.059)	(0.020)
Very satisfied or satisfied with current job	**−0.004**	**−0.001**	**−0.008**
	(0.007)	**(0.011)**	**(0.010)**
Indifferent about current job	**0.014**	**0.084**	**−0.094**
	(0.053)	**(0.053)**	**(0.077)**
Very satisfied with pay	0.173***	0.179***	0.160***
	(0.015)	(0.021)	(0.017)
Satisfied with pay	0.109***	0.106***	0.106***
	(0.014)	(0.020)	(0.015)
Dissatisfied with pay	0.049***	0.053***	0.043***
	(0.013)	(0.019)	(0.015)
Indifferent about pay	0.005	−0.002	0.016
	(0.051)	(0.065)	(0.080)
Full time	0.057***	0.100***	0.032***
	(0.006)	(0.012)	(0.006)
Member of union or collective bargaining agreement	0.102***	0.109***	0.097***
	(0.006)	(0.008)	(0.007)
No. of observations	70,345	40,230	30,115
Adjusted R-squared	0.208	0.177	0.243

NOTES: Standard errors are in parentheses. *** indicates statistically significant at 1% ** indicates statistically significant at 5%, and * indicates statistically significant at 10%. All coefficients are estimated using sampling weights provided by Statistics Canada and all the standard errors are computed using 100 sets of bootstrap weights provided by Statistics Canada. The dependent variable is estimated firm effects after controlling for workers and job characteristics (estimates of ψ from equation (3)). The regressions also include control for age (eight categories), quartic in experience and quadratic in years of seniority, occupation (five categories), number of children (four categories), immigration status and ethnicity (four categories).

TABLE 4
Firm-specific premiums and firm average characteristics

Industry		
Resource	0.150***	(0.020)
Labour intensive tertiary manufacturing	−0.117***	(0.020)
Secondary product manufacturing	0.044***	(0.015)
Capital intensive tertiary manufacturing	0.013	(0.017)
Construction	0.157***	(0.020)
Transportation, warehousing, wholesale	−0.012	(0.017)
Communication and other utilities	0.064***	(0.016)
Retail trade and consumer services	−0.156***	(0.017)
Finance and insurance	0.050**	(0.021)
Real estate, rental and leasing operations	−0.053*	(0.031)
Business services	0.072***	(0.023)
Education and health services	0.109***	(0.028)
Information and cultural industries	0.036*	(0.020)
No. of employees		
20–99	0.027**	(0.010)
100–499	0.081***	(0.013)
500 or more	0.122***	(0.031)
No. of competing firms		
0	−0.025	(0.025)
1–5	−0.066***	(0.018)
6–20	−0.047***	(0.016)
Foreign ownership		
1%–49%	−0.008	(0.027)
50%–90%	−0.104	(0.064)
90%–100%	0.028	(0.018)
Existence of any employment equity program	0.007	(0.027)
Existence of any pay equity program	0.052*	(0.028)
Proportion of full-time workers	0.142***	(0.032)
Proportion of workers covered by a collective bargaining agreement	0.015	(0.023)
Good rating of labour-management relations	−0.009	(0.012)
Existence of any innovative work practices	−0.033**	(0.015)
Quit rate	−0.171***	(0.063)
Log of value added per worker (proxy for productivity)	0.066***	(0.007)
Log of training expenditures per worker	0.006***	(0.002)
Z-score measure for provision of different benefits	0.006***	(0.001)
Existence of incentive schemes in the compensation system	0.069***	(0.014)
No. of observations	14,015	
Adjusted R-squared	0.416	

NOTES: Standard errors are in parentheses. *** indicates statistically significant at 1%, ** indicates statistically significant at 5% and * indicates statistically significant at 10%. All coefficients are estimated using sampling weights provided by Statistics Canada and all the standard errors are computed using 100 sets of bootstrap weights provided by Statistics Canada. The dependent variable is estimated firm effects after controlling for workers and job characteristics (estimates of ψ from equation (3)). For a detailed description of variables included in the regression please refer to the technical appendix, available at cje.economics.ca.

gender differences in sorting. It seems difficult to think of supply-side reasons that are able to explain why women would avoid jobs with efficiency wages. On the other hand, as Bulow and Summers (1986) point out, since a central element in efficiency wage theory is wage differences that are unrelated to productivity differences across workers, it is reasonable to think that it can provide the basis for a theory of discrimination.[24]

Firms paying efficiency wages (or higher efficiency wages) would believe it is not optimal to hire women if they assume that females do not alter their behaviour in response to higher wages, for instance because they are less career orienteed or less productive, or if they assume it takes a higher wage increase to change women's behaviour such as shirking because they have higher quit rates so the cost of losing their job is less than the cost for men. These assumptions could be based on stereotypes about female workers, considering them more communal, caring and family oriented compared to males who might be considered more work oriented. If stereotype-based assumptions about females are incorrect, these lower wages are discriminatory and inefficient.[25]

I adopt an empirical test proposed by Hellerstein et al. (2002) to examine the presence of discrimination against women in the labour market in the context of efficiency wage theory, which would limit females' access to high-paying firms. I implement the test by examining the cross-sectional relationship between a firm's profitability and its gender composition.[26] In the absence of discrimination against females that blocks their access to high-paying firms, females' under-representation in high-paying firms and the gender wage gap generated through this sorting must reflect only observed or unobserved gender differences in productivity or preferences. Therefore, we should not observe any systematic relationship between profitability and within-firm proportion of females. A finding that firms with a higher proportion of females earn higher profits, in contrast, would be consistent with sex discrimination against females that limits their access to high-paying firms (firm that pay efficiency wages or higher efficiency wages).

The following example might make the intuition behind this conclusion clearer. Imagine a labour market with efficiency wages where there are two firms. Firm A is a high-paying firm, and firm B is a low-paying firm. There are 10 male and 10 female workers in the labour market. Females have the same distribution of productivity as males, but within each group workers have different productivity levels. Firm A has a discriminatory taste against employing females and therefore

24 For example, Yellen (1984) argues that in the context of the efficiency wage model, employers can costlessly discriminate against a group of workers with some observable characteristics. Bulow and Summers (1986) also develop an efficiency wage model that rationalizes discrimination based on group differences that are unrelated to productivity.

25 Even if these assumptions about females are on average correct, it does not rule out the possibility of statistical discrimination against females that could lead to gender differences in sorting across firms. For instance, there could be statistical discrimination against women based on beliefs in gender differences in turnover rates that would lead to inter-firm gender segregation (Aldrich and Buchele 1989).

26 Since the question regarding the total number of females in the firm was included in the WES starting 2003, the regression is implemented using only pooled 2003 and 2005 firm-level data.

hires mainly males, while firm B has no prejudice against females. Therefore, females are under-represented in firm A and over-represented in firm B. Since firm A discriminates against employing females, there will be some male workers employed at firm A who are less productive than some of the females employed at firm B but who receive higher wages than those higher-productivity females at firm B, based solely on their affiliation with firm A. As a result, although firm A will have on average workers with higher productivity than firm B (because it pays higher wages and attracts better workers, on average), the difference in average wages between firm A and firm B will be larger than the difference in average productivity between the two firms. Therefore, firm B will have a higher profit than firm A.

The results are reported in table 5 and are robust across different specifications. They suggest that irrespective of the set of control variables included in the regression, there is a positive relationship between the proportion of females in the workforce and firms' profitability. For example, the point estimate in column (6) indicates that a 10 percentage point increase in female employment in an average firm increases the profit rate by 0.73 percentage point.[27] These results are consistent with static implication of Becker's model of employer discrimination.[28] Altogether, my results suggest that gender differences in preferences or productivity cannot explain inter-firm gender segregation and therefore the glass ceiling faced by females. My results, however, favour the predictions of efficiency wage theory with higher-paying firms discriminating against females.

5. Conclusion

I find clear evidence that females face economy-wide glass ceilings that are driven mainly by glass doors (i.e., segregation of females in lower-paying firms). I find similar patterns for different subgroups of the workforce (i.e., females without any dependent child and all males, females with at least one dependent child and all males, single workers without any dependent children, workers with a bachelor's degree or higher, workers without a bachelor's degree, workers younger than 44 and workers older than 43). I also find that females experience substantial wage gaps throughout the conditional wage distribution, even within firms, with their male counterparts. I find no evidence that these wage patterns can be explained

27 These results are consistent with Hellerstein et al. (2002).

28 Dynamic implication of Becker's employer discrimination model suggests that in a competitive environment this difference in profitability might eliminate discrimination. However, as Hellerstein et al. (2002) also argue, this conclusion is not always correct as it depends on several factors, such as the nature of the product market, barriers to entry, transferability of assets and the type of discrimination. Those who argue that employer discrimination cannot persist usually gloss over the conditions that need to be satisfied. For instance, the difference in profitability will not drive the discriminatory employers out of the market if the competition in the product market is not strenuous enough. Ederington and Sanford (2012) develop a theoretical model in which they identify conditions under which discriminatory firms will survive in the long run, even within Becker's framework.

TABLE 5
Firm's profitability and the sex composition of the workforce

% females	0.086***	0.101***	0.089***	0.075***	0.074***	0.073***
	(0.024)	(0.024)	(0.024)	(0.026)	(0.026)	(0.026)
% with bachelor's degree or higher	0.000	−0.010	−0.027	−0.031	−0.029	−0.029
	(0.024)	(0.023)	(0.025)	(0.024)	(0.025)	(0.025)
% with college degree	−0.023	−0.025	−0.038	−0.043*	−0.043*	−0.043*
	(0.023)	(0.023)	(0.024)	(0.024)	(0.024)	(0.024)
% with high school diploma	0.018	0.017	0.005	0.001	0.002	0.002
	(0.023)	(0.022)	(0.026)	(0.023)	(0.023)	(0.023)
% married	−0.007	−0.012	−0.014	−0.015	−0.016	−0.016
	(0.021)	(0.020)	(0.019)	(0.020)	(0.019)	(0.019)
% immigrant	−0.006	−0.007	−0.007	−0.005	−0.003	−0.001
	(0.024)	(0.024)	(0.022)	(0.021)	(0.020)	(0.020)
% with at least 1 dependent child	0.021	0.022	0.019	0.020	0.019	0.018
	(0.019)	(0.018)	(0.019)	(0.018)	(0.018)	(0.018)
% between 35 and 54	−0.022	−0.029	−0.028	−0.032	−0.032	−0.033
	(0.023)	(0.022)	(0.022)	(0.021)	(0.020)	(0.021)
% older than 54	−0.020	−0.021	−0.020	−0.022	−0.024	−0.028
	(0.039)	(0.039)	(0.041)	(0.038)	(0.037)	(0.038)
% with 10 to 25 years of exp.	−0.006	−0.016	−0.017	−0.016	−0.016	−0.014
	(0.026)	(0.026)	(0.026)	(0.024)	(0.024)	(0.024)
% with > 25 years of experience	0.028	0.010	0.012	0.018	0.020	0.024
	(0.030)	(0.031)	(0.031)	(0.030)	(0.030)	(0.030)
% unionized		−0.041**	−0.043**	−0.045**	−0.040**	−0.040**
		(0.019)	(0.018)	(0.018)	(0.020)	(0.019)
% full time		0.038*	0.030	0.033	0.034	0.036*
		(0.022)	(0.020)	(0.021)	(0.021)	(0.021)
% with 10 to 25 years of tenure		0.036*	0.034	0.033*	0.032*	0.031
		(0.021)	(0.022)	(0.019)	(0.019)	(0.019)
% with > 25 years of tenure		0.048*	0.048*	0.047*	0.047*	0.045*
		(0.026)	(0.026)	(0.025)	(0.025)	(0.025)
Control for % with different occupations	NO	NO	YES	YES	YES	YES
Control for industry	NO	NO	NO	YES	YES	YES
Control for firm size	NO	NO	NO	NO	YES	YES
Control for degree of competition	NO	NO	NO	NO	NO	YES
No. of observations	6,089	6,089	6,089	6,089	6,089	6,089

NOTES: Standard errors are in parentheses. *** indicates statistically significant at 1%, ** indicates statistically significant at 5% and * indicates statistically significant at 10%. All coefficients are estimated using sampling weights provided by Statistics Canada and all the standard errors are computed using 100 sets of bootstrap weights provided by Statistics Canada. The dependent variable is profit rate: (gross operating revenue less gross operating income) / gross operating income. Observations with extreme values on the dependent variable (around 1% of the sample) are deleted. The regressions are based only on 2003 and 2005 observations since the actual number of females in a firm was reported only for those years. Control for industry includes 14 categories. Control for firm size includes four categories (fewer than 20 employees, 20 to 99, 100 to 499, more than 500). Control for degree of competition is based on the number of competing firms and includes four categories (0, 1 to 5, 6 to 20 and more than 20). Control for % with different occupations includes six occupational categories.

by compensating differentials. However, my results support the predictions of the efficiency wage theory with higher-paying firms discriminating against females.

The policy implications of my results are two-fold. First, policies that aim to identify and eliminate the glass ceiling faced by females will be more effective if they focus on mechanisms that segregate women into lower-paying firms. This is possible, for instance, by addressing hiring practices in higher-paying firms. Employment equity policies that target employment decisions directly could be effective in this context and will have a bigger effect in reducing the gender wage gap in higher-paying jobs. Second, more general policies that try to improve females' labour market outcomes throughout the wage distribution will be more effective if they focus on practices that improve females' labour market performance within firms. To the extent that the jobs performed by males and females within firms require similar skills, effort, responsibilities and working conditions, policies that target pay decisions directly, such as pay equity programs, or policies that promote gender equality in promotion opportunities within firms could be effective in reducing the within-firm gender wage gap. Finally, the sizable within-firm gender wage gap found in my analysis throughout the wage distribution suggests that further research is required to identify the sources of within-firm gender wage gap.

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